

Technical Design Report: Astable Multivibrator LED Flasher PCB

1. Executive Summary

This document outlines the design, schematic capture, and PCB layout of an LED flasher circuit developed using KiCad software. The system leverages a 555 timer integrated circuit configured in an astable mode to autonomously generate a continuous oscillating output. This pulsed signal drives a Light Emitting Diode (LED), establishing a reliable, time-based visual indicator suitable for signaling, decorative lighting, and educational demonstrations.

2. Circuit Architecture and Theory of Operation

The core of the flasher circuit is the 555 timer IC, which regulates the timing of the LED flash through its internal comparator and flip-flop circuitry. By operating in astable mode, the timer's output continuously toggles between high and low states without requiring an external trigger.

The oscillation frequency and duty cycle are strictly governed by an external passive RC network consisting of two resistors and one capacitor. Specifically, the charging and discharging intervals are set by R_1 , R_2 , and C_1 , which dictate the exact ON and OFF durations of the output LED. A current-limiting resistor (R_3) is placed in series with the LED to ensure safe operation and prevent thermal damage from excess current.

$$f = 1.44 / ((R_1 + 2R_2) \times C_1)$$

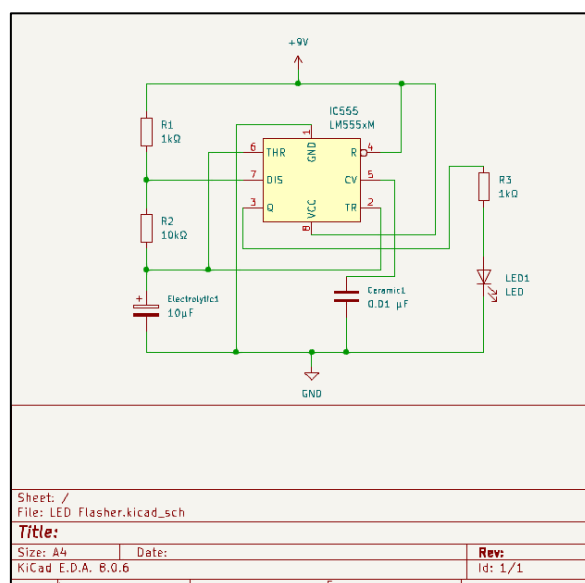


Figure 1: KiCad schematic detailing the LM555xM astable configuration and passive timing network.

3. Component Selection and Footprint Allocation

The PCB design utilizes a hybrid component packaging strategy, combining a Surface Mount Device (SMD) controller with Through-Hole Technology (THT) passives to balance footprint density with assembly ease.

The exact Bill of Materials (BOM) and footprint assignments implemented in the KiCad footprint editor are as follows:

- **Integrated Circuit (IC555):** LM555xM timer assigned to an SMD SOIC-8 package (3.9×4.9 mm, 1.27 mm pitch).
- **Timing Resistors (R_1, R_2):** 1 k Ω and 10 k Ω values assigned to standard horizontal axial footprints (R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm).
- **Current-Limiting Resistor (R_3):** 1 k Ω assigned to the identical axial footprint.
- **Timing Capacitor (C_1):** 10 μ F electrolytic capacitor utilizing a polarized radial footprint (CP_Radial_D5.0mm_P2.50mm).
- **Decoupling Capacitor (C_2):** 0.01 μ F ceramic capacitor utilizing a disc footprint (C_Disc_D3.0mm_W2.0mm_P2.50mm).
- **Output (LED1):** Standard red LED utilizing a 3.0 mm through-hole footprint.
- **Power Source:** 9V battery (external; no direct board footprint required other than terminal access).

Symbol : Footprint Assignments		
1 Ceramic1 -	0.01 μ F :	Capacitor_THT:C_Disc_D3.0mm_W2.0mm_P2.50mm
2 Electrolytic1 -	10 μ F :	Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
3 IC555 -	LM555xM :	Package_SO:SOIC-8_3.9x4.9mm_P1.27mm
4 LED1 -	LED :	LED_THT:LED_D3.0mm
5 R1 -	1k Ω :	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
6 R2 -	10k Ω :	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
7 R3 -	1k Ω :	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal

Figure 2: Footprint assignment table mapping schematic symbols to physical KiCad packages.

4. Printed Circuit Board (PCB) Layout

Following schematic validation, the netlist was imported into the KiCad PCB editor for physical routing. The components were spatially arranged on the canvas to minimize trace length between the timing capacitor, resistors, and the threshold/discharge pins of the LM555xM IC. Tracks were routed to ensure secure electrical connections while adhering to standard clearance and trace-width manufacturing constraints.

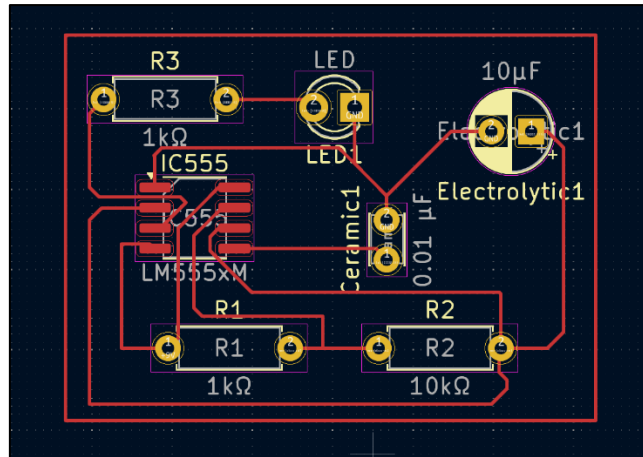


Figure 3: Copper trace routing and component placement in the KiCad PCB editor.

To verify the mechanical clearances, component alignment, and overall board dimensions prior to manufacturing, the fully routed layout was rendered in KiCad's 3D view mode.

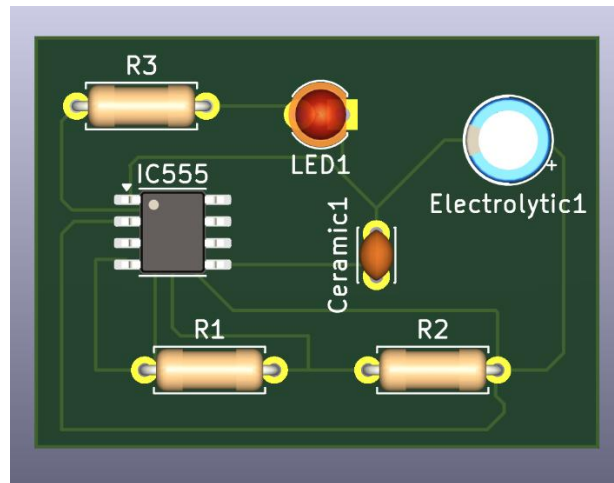


Figure 4: 3D mechanical rendering of the fully assembled LED Flasher PCB.

5. Practical Applications

The finished astable multivibrator circuit offers high reliability and scalability for multiple use cases:

- **Signaling Applications:** Integration into visual alert systems, warning beacons, and status indicators where attention-grabbing intermittent light is required.
- **Decorative Lighting:** Deployment in hobbyist projects, displays, and consumer electronics to provide rhythmic visual appeal.
- **Educational Training:** Serving as a foundational hardware module to demonstrate RC time constants, continuous oscillation, and baseline PCB layout principles.